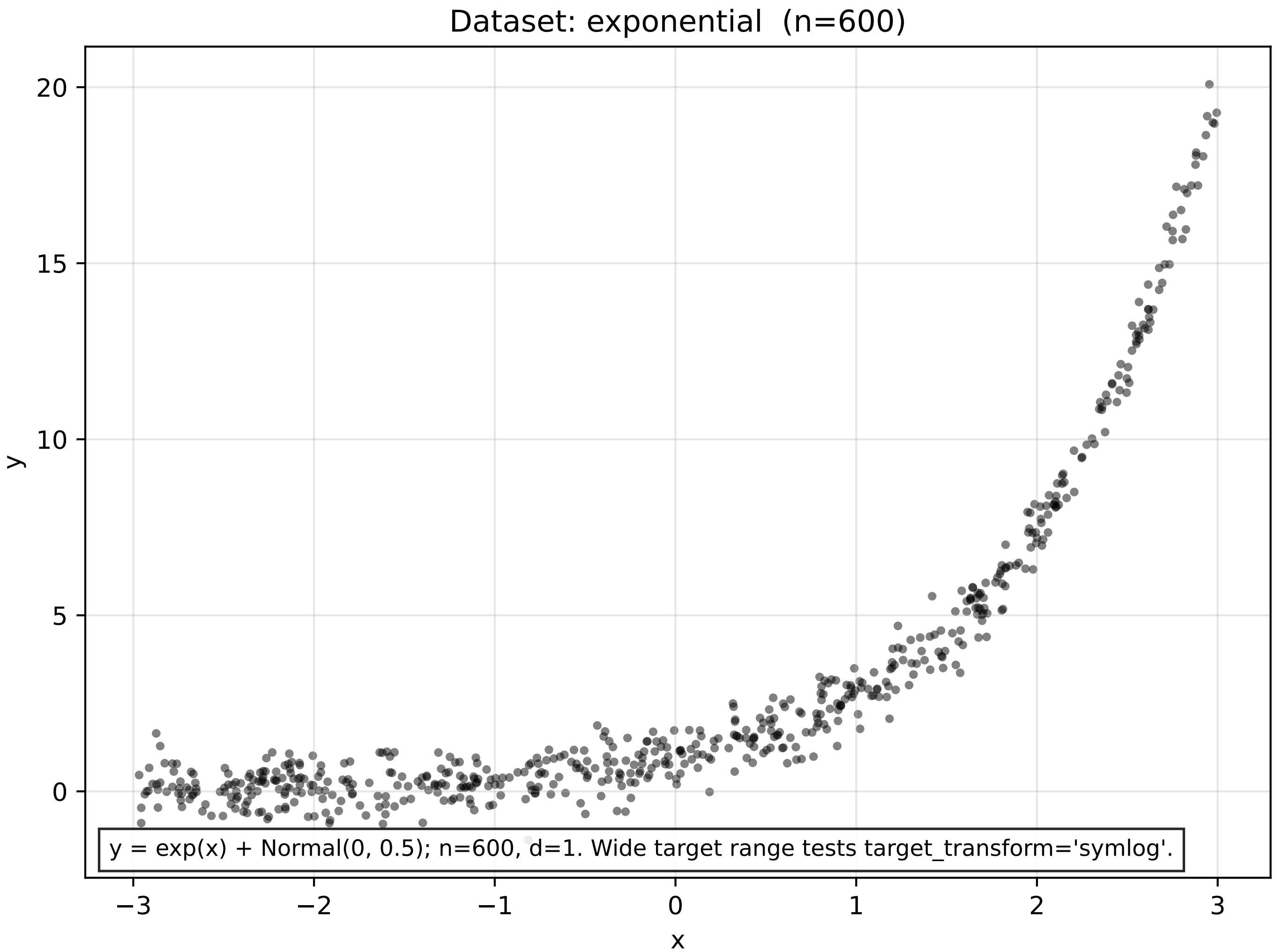
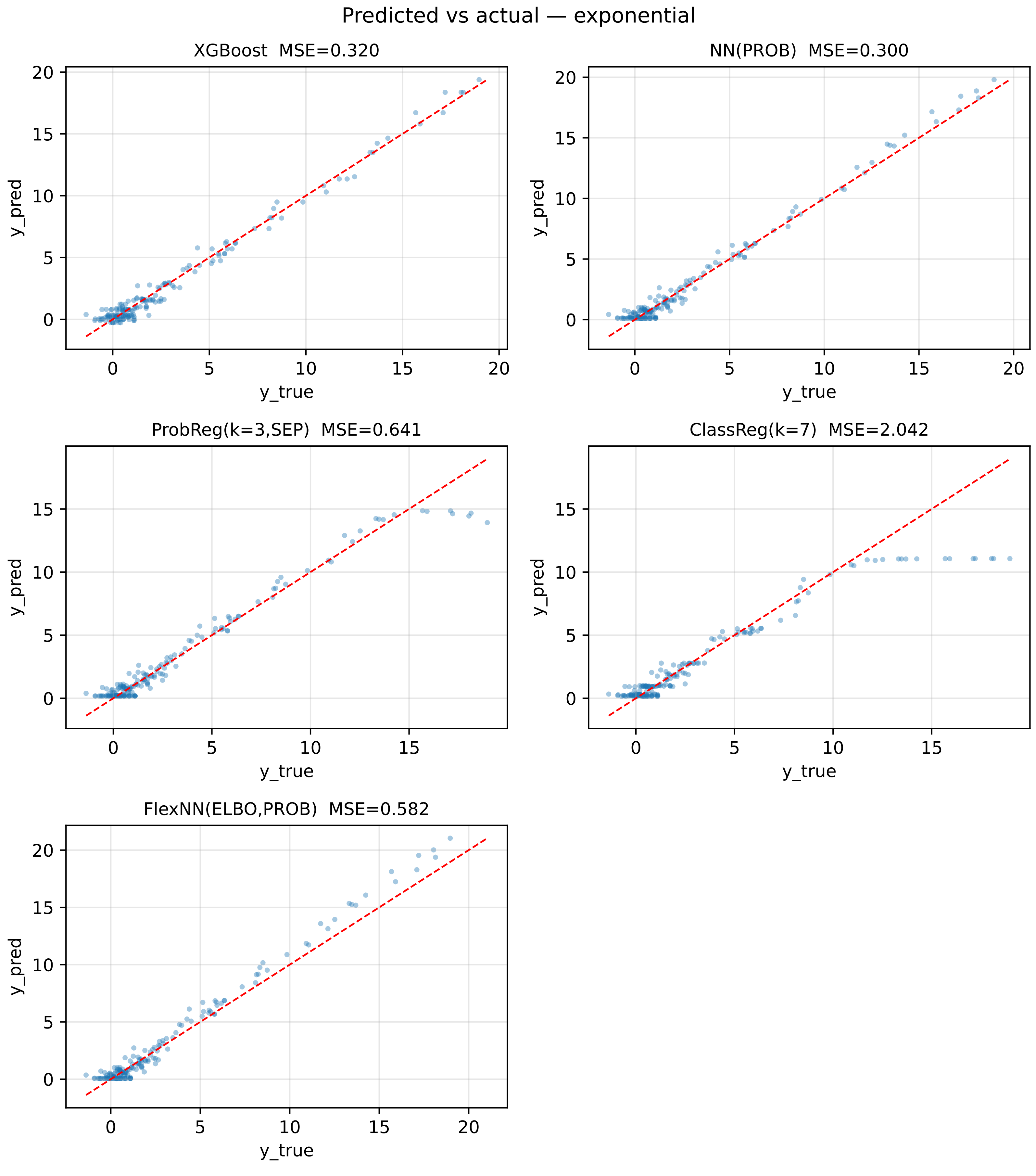
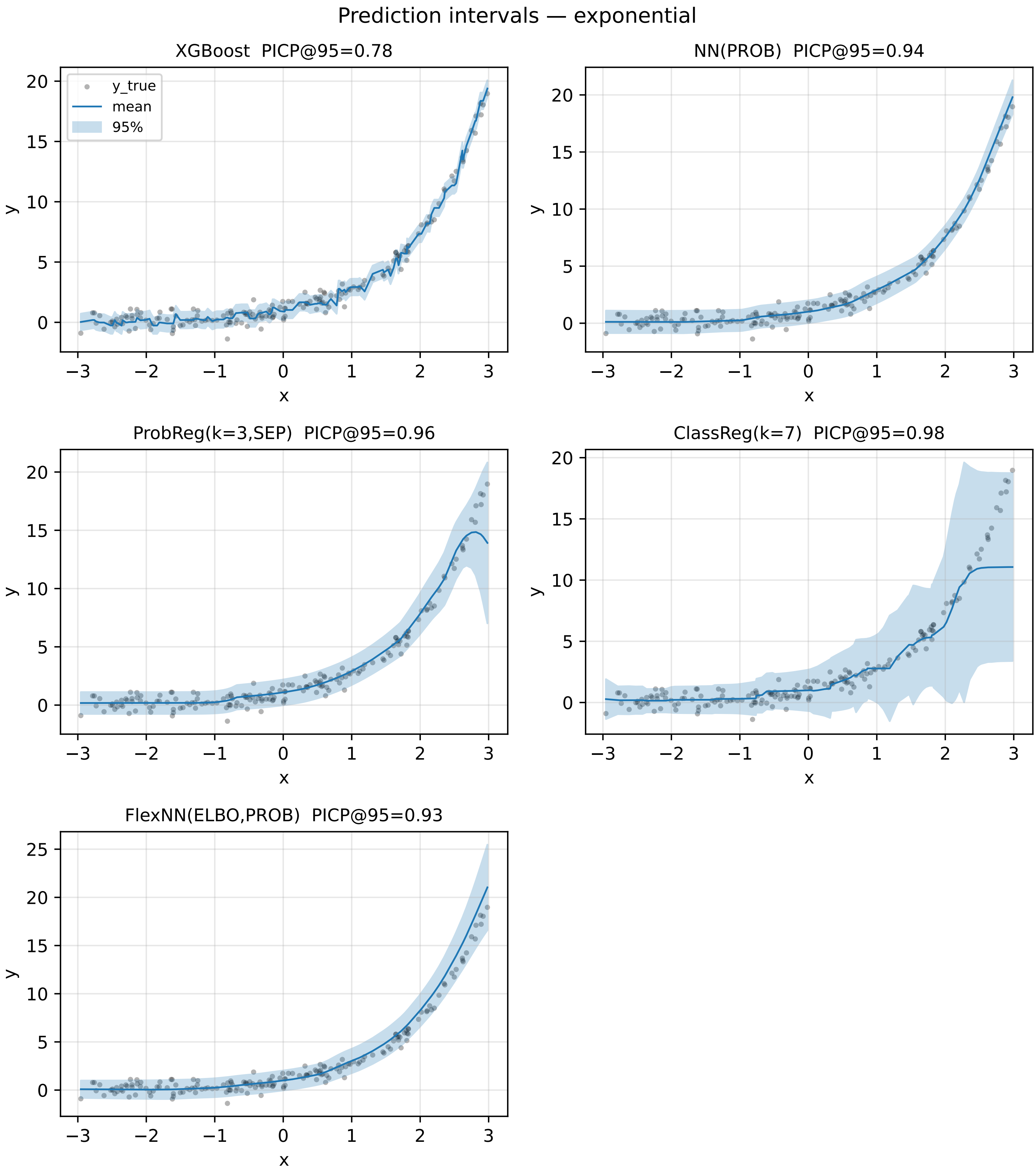

Dataset: exponential (n=600)
20
15
10
5
0
y
−3
−2
−1
0
1
2
3
x
y = exp(x) + Normal(0, 0.5); n=600, d=1. Wide target range tests target_transform='symlog'.

Model comparison

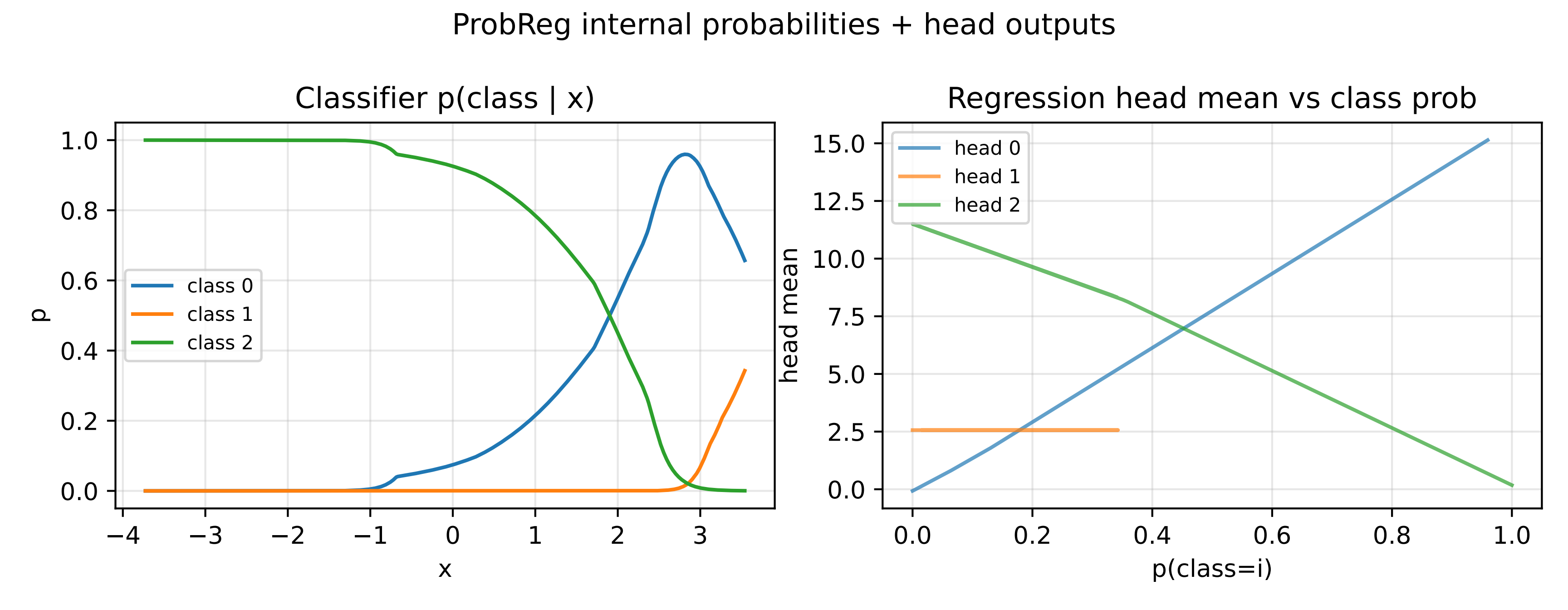
model	mse	nll	crps	picp95	mpiw95	sharpness
XGBoost	0.3198	1.1694	0.3328	0.7833	1.3763	0.3511
NN(PROB)	0.2997	0.8075	0.3049	0.9389	2.0689	0.5278
ProbReg(k=3,	0.6411	0.9075	0.3645	0.9556	2.6615	0.6790
ClassReg(k=7	2.0421	1.2917	0.5943	0.9778	5.4795	1.3979
FlexNN(ELBO,	0.5824	0.9699	0.3924	0.9278	2.5727	0.6563



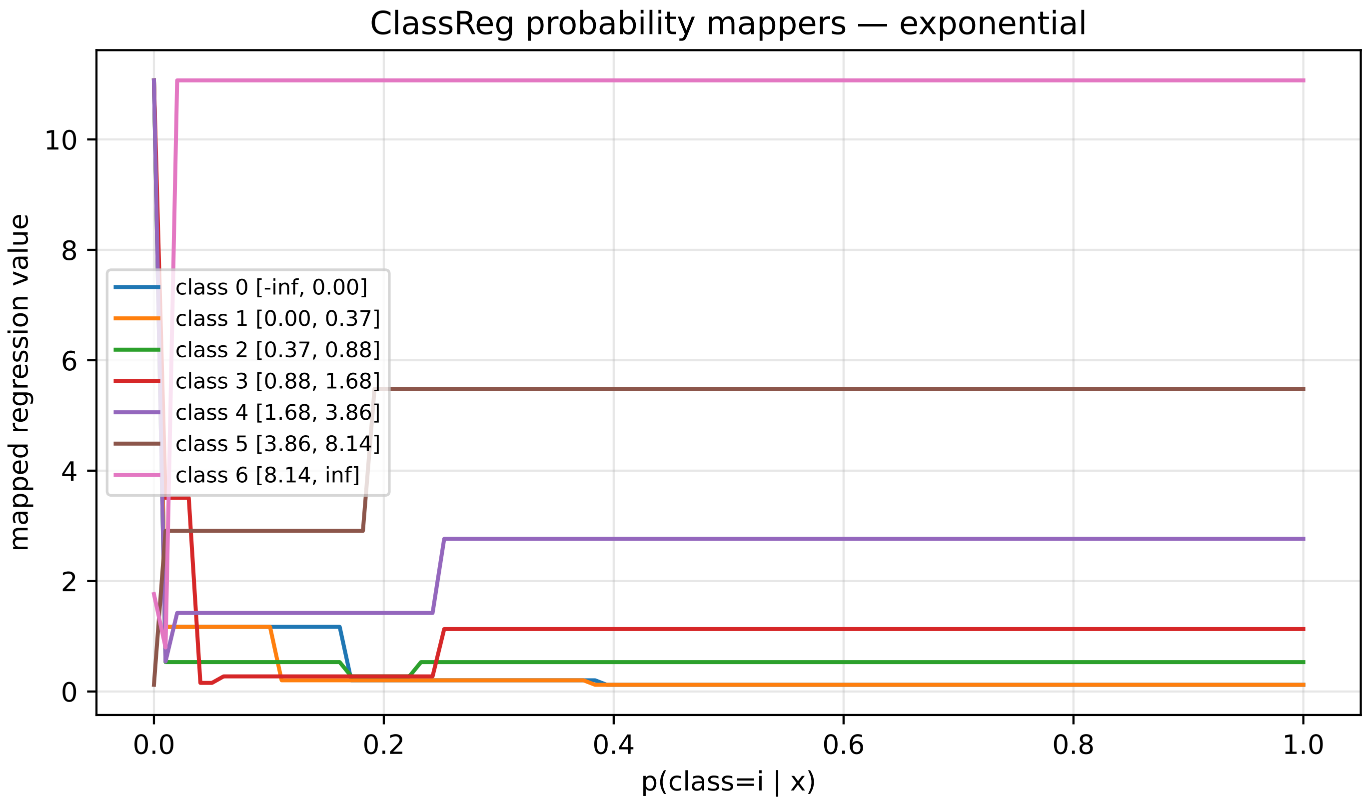
Predicted vs actual — exponential
XGBoost MSE=0.320
NN(PROB) MSE=0.300
ProbReg(k=3,SEP) MSE=0.641
ClassReg(k=7) MSE=2.042
FlexNN(ELBO,PROB) MSE=0.582
y_pred
y_true

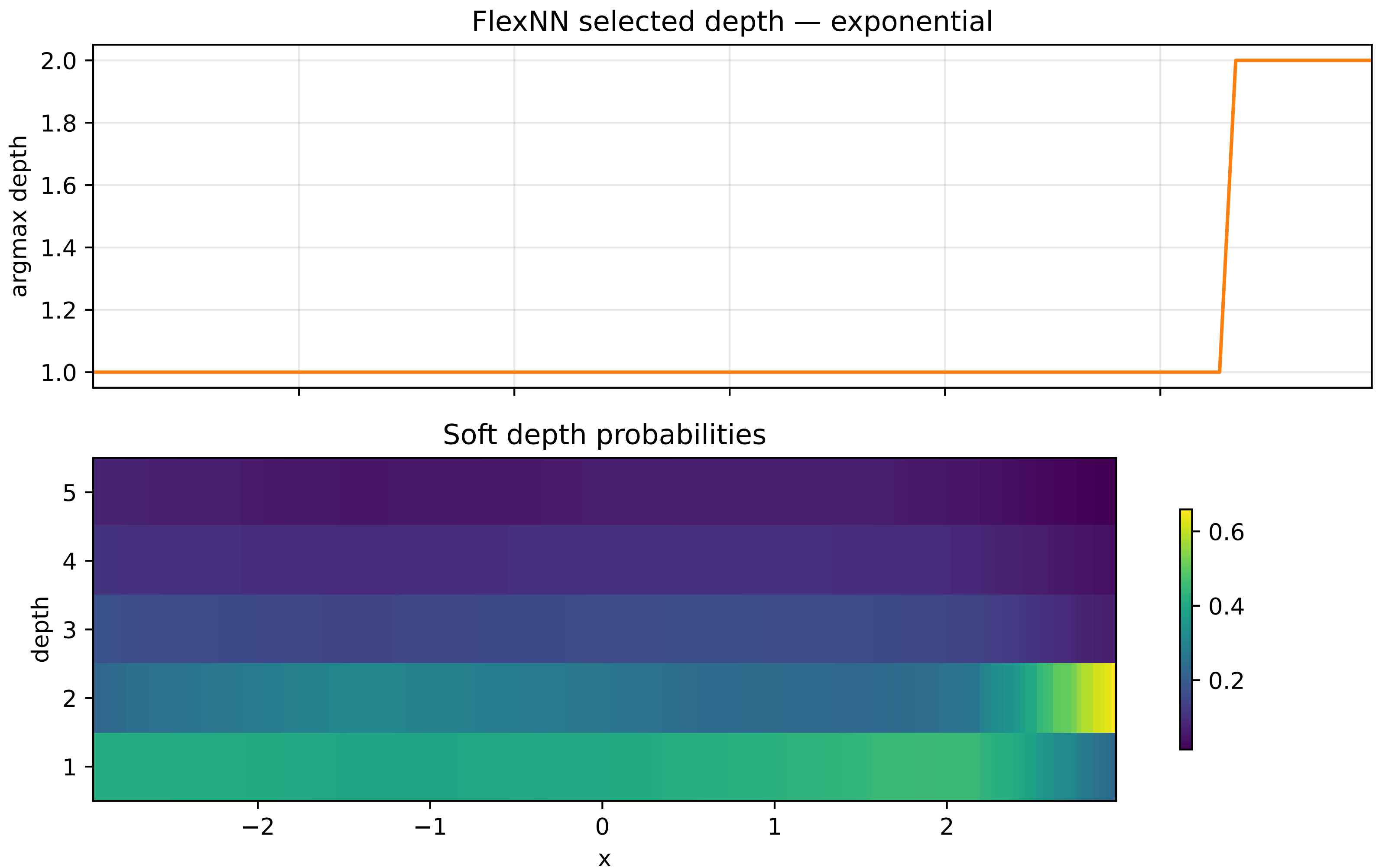


Prediction intervals — exponential
XGBoost PICP@95=0.78
NN(PROB) PICP@95=0.94
ProbReg(k=3,SEP) PICP@95=0.96
ClassReg(k=7) PICP@95=0.98
FlexNN(ELBO,PROB) PICP@95=0.93
y_true
mean
95%
x
y



ProbReg internal probabilities + head outputs
Classifier p(class | x)
class 0
class 1
class 2
p
x
1.0
0.8
0.6
0.4
0.2
0.0
−4
−3
−2
−1
0
1
2
3
Regression head mean vs class prob
head 0
head 1
head 2
head mean
p(class=i)
15.0
12.5
10.0
7.5
5.0
2.5
0.0
0.2
0.4
0.6
0.8
1.0


ClassReg probability mappers — exponential
mapped regression value
p(class=i | x)
0
2
4
6
8
10
0.0
0.2
0.4
0.6
0.8
1.0
class 0 [-inf, 0.00]
class 1 [0.00, 0.37]
class 2 [0.37, 0.88]
class 3 [0.88, 1.68]
class 4 [1.68, 3.86]
class 5 [3.86, 8.14]
class 6 [8.14, inf]



FlexNN selected depth — exponential
argmax depth
2.0
1.8
1.6
1.4
1.2
1.0
Soft depth probabilities
depth
5
4
3
2
1
−2
−1
0
1
2
x
0.6
0.4
0.2

```
Ranked MSE for exponential:
  - NN(PROB): MSE=0.2997  NLL=0.808  PICP95=0.94
  - XGBoost: MSE=0.3198  NLL=1.169  PICP95=0.78
  - FlexNN(ELBO,PROB): MSE=0.5824  NLL=0.970  PICP95=0.93
  - ProbReg(k=3,SEP): MSE=0.6411  NLL=0.907  PICP95=0.96
  - ClassReg(k=7): MSE=2.0421  NLL=1.292  PICP95=0.98

Notes:
Exponential targets span ~20x; symlog transform and probabilistic uncertainty should both be bene
```